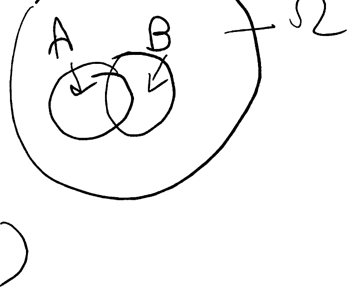
PRST Class (16/09/2020)

$$Y = a + bX$$
$$\bar{Y} = a + b\bar{X}$$
$$V(Y) = b^2 V(x).$$

Result: If A and B are ind, then

(i) $P(A \cap B^c) = P(A) \cdot P(B^c)$

(ii) $P(A^c \cap B^c) = P(A^c)\, P(B^c)$



proof: (i). $P(A \cap B) = P(A)\, P(B)$.

$$P(A \cap B^c) = P(A) - P(A \cap B)$$
$$= P(A) - P(A)\,P(B)$$
$$= P(A)\,(1 - P(B))$$
$$= P(A) \times P(B^c).$$

(ii)
$$P(A^c \cap B^c) = P((A \cup B)^c)$$
$$= 1 - (P(A) + P(B) - P(A \cap B))$$
$$= 1 - P(A) - P(B) + P(A) \cdot P(B)$$
$$= 1 - P(A) - P(B)\,(1 - P(A))$$
$$= P(A^c) \times P(B^c).$$

Ex

$$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$$
$$P(\omega_i) = \frac{1}{4} \quad i = 1, 2, 3, 4$$
$$A_1 = \{\omega_1, \omega_2\}, \quad A_2 = \{\omega_1, \omega_3\}$$
$$A_3 = \{\omega_1, \omega_4\}$$
$$P(A_i) = \frac{1}{2} \quad i = 1, 2, 3$$
$$P(A_1 \cap A_2) = \frac{1}{4} = P(A_1)P(A_2)$$
$$P(A_1 \cap A_3) = \frac{1}{4} = P(A_1)P(A_3)$$
$$P(A_2 \cap A_3) = \frac{1}{4} = P(A_2) \times P(A_3).$$
$$P(A_1 \cap A_2 \cap A_3) = \frac{1}{4}$$
$$P(A_1)P(A_2)P(A_3) = \frac{1}{8}$$

$A_1, \ldots, A_n$

$$P(A_{i_1} \cap A_{i_2}) = P(A_{i_1})P(A_{i_2})$$
$$P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) = P(A_{i_1})P(A_{i_2})P(A_{i_3})$$
$$\vdots$$
for $2 \le k \le n$
$$P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k})$$

mutually independent

$P(A_1 \cap A_2 \cap A_3)$

$P(A_1)P(A_2)P(A_3) = \frac{1}{8}$

not mutually ind.

for $2 \leq k = ...$

$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k})$

$= P(A_{i_1})P(A_{i_2}) \dots P(A_{i_k})$

Ex 1. What is the prob that all 3 children in a family have different birthdays?

— $P = \frac{365 \times 364 \times 363}{365 \times 365 \times 365} = 0.99.$

Ex 2. A batch of products contain 10 products of which 4 are defective. If 3 products are chosen at random, what is the prob that none of them will be defective?

$\rightarrow \frac{{}^6C_3}{{}^{10}C_3} = \frac{1}{6}$

Ex. A coin is tossed 7 times. find the prob that we will see at least 2 Heads;

where $P(H) = P$ $\quad 0 < P < 1.$

HTT... T

THT... T

$\rightarrow 1 - P(\text{at most } 1\ H).$

$= 1 - P(\text{no } H \text{ or only } 1\ H)$

$= 1 - \left[{}^7C_0\, p^0 (1-p)^7 + {}^7C_1\, p^1 (1-p)^6 \right]$

Ex Two players A an B throw a die alternatively. He who first gets a six will win. If A begins, what is the prob that A will win?

— E – getting six

, getting sixe

— E – getting six
F – not getting six

E → Prob = $\frac{1}{6}$

FFE Prob = $\left(\frac{5}{6}\right)^2 \frac{1}{6}$

FFFFE Prob = $\left(\frac{5}{6}\right)^4 \frac{1}{6}$

⋮

$$P(A \text{ will win}) = \frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \cdots$$

$$= \frac{\frac{1}{6}}{1 - \left(\frac{5}{6}\right)^2} = \frac{6}{11}$$

Random variable.

ANANAA……

$X : \Omega \longrightarrow \mathbb{R}$.

Random variable is a function fr sample space Ω to real line.